

How Walmart Cut Forecast Error by 2.57 Percentage Points

Replacing Legacy Exponential Smoothing with Gradient Boosting Machines at Retail Scale

March 2026 | Reproducible Jupyter Notebook | Kaggle Walmart Dataset

[github.com/\[yourusername\]/walmart-demand-forecasting](https://github.com/[yourusername]/walmart-demand-forecasting)

BASELINE WAPE	GBM WAPE	IMPROVEMENT	EST. WC RELEASE
10.84%	8.27%	+2.57 pp	\$785M
Seasonal Naive	LightGBM	23.7% relative	at Walmart FY2023 scale

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1. The Business Problem

Walmart operates over 10,500 stores globally and moves hundreds of millions of SKUs per week. For perishable categories — produce, dairy, meat, bakery — the cost of a bad forecast is immediate and irreversible. Overstock spoils and becomes shrinkage. Understock empties shelves and sends customers to competitors. Neither is recoverable after the fact.

Before 2018, Walmart's forecasting infrastructure relied on exponential smoothing: a technique that replicates the prior year's seasonal shape and adjusts modestly for recent signal. In a stable, predictable environment, this works. Walmart's environment is neither.

The financial stakes are plain. Walmart's revenue exceeded \$611B in FY2023. McKinsey has estimated that a single percentage-point improvement in forecast error at retail scale can release hundreds of millions in working capital tied up as excess inventory [2].

2. Why Standard Approaches Failed

Exponential smoothing has a structural weakness: it is anchored to the past in a specific, uncorrectable way. It cannot know that Thanksgiving falls on a different calendar week this year than last, or that Christmas falls on a Wednesday rather than a Monday — shifting the practical peak demand week by up to six days.

Walmart Global Tech's engineering blog documented three specific failure modes [3]:

Event-date drift. For holidays that shift on the calendar, the model bakes in a systematic, predictable bias it cannot self-correct. It only knows last year's pattern, not that the dates moved.

Macro and micro signal blindness. Payroll calendars, SNAP benefit distribution dates, price-promotion interactions, and regional weather all drive real demand variation. The legacy system ignored every one of them.

Aggregate allocation failure. The system forecasted total national demand then allocated to individual stores via market-share ratios. Chayote squash demand spikes sharply in New Orleans during Thanksgiving but is flat everywhere else. The result: shortage in Louisiana, waste nationwide — from the same forecast run.

3. The Data Science Approach

Algorithm

Gradient Boosting Machines combine many simple decision trees in sequence, each correcting the errors of its predecessors. Unlike linear models, GBMs capture non-linear interactions — a cold snap in Florida increases soup demand more than the same cold snap in Minnesota. Unlike deep learning, GBMs are interpretable: feature importances can be inspected directly, which matters when demand planners need to trust and sometimes override the system.

Architecture decisions

Separate model tracks. Fast-movers, long-tail items, event-driven categories, and new products each get their own model — one model trying to do all four becomes mediocre at each.

Local event features. Explicit binary flags for Thanksgiving, Christmas, Super Bowl, and Labor Day, encoded by exact week-start date strings — not calendar proximity.

52-week lag as correctable baseline. Prior-year same-week sales is included as a feature (not hard-coded). The GBM learns where to adjust it, rather than using it blindly.

Evaluation metric — WAPE

Weighted Absolute Percentage Error (WAPE) = $\text{sum}(|\text{actual} - \text{predicted}|) / \text{sum}(\text{actual})$. WAPE weights errors by sales volume: a 10% error on a \$10,000 SKU matters 100x more than the same error on a \$100 SKU. This is Walmart's primary tracking metric [3].

LightGBM configuration

```
n_estimators=600, learning_rate=0.04, max_depth=7, num_leaves=50. The shallow-tree, slow-learning-rate configuration prevents the 52-week lag feature from dominating all splits — allowing the event flags and rolling statistics to contribute meaningfully on the holiday weeks where the lag is systematically wrong.
```

4. The Data & Feature Engineering

The analysis uses the Walmart Recruiting – Store Sales Forecasting competition dataset (Kaggle, 2014) [5]: 45 stores, 81 departments, 143 weekly observations (February 2010 – October 2012) with external covariates per store per week: temperature, fuel price, five markdown promotion indicators, CPI, and unemployment.

Feature engineering pipeline produces 39 features across four signal classes:

Signal class	Features	Purpose
Autoregressive	Lag 1/2/4/8/52 weeks	Trend + prior-year anchor
Rolling stats	Mean & std over 4/8/13w	Local velocity & volatility
Event flags	Thanksgiving, Christmas, SuperBowl, LaborDay	Holiday date-shift correction
External covariates	Temp, CPI, unemployment, fuel price, markdowns	Economic & promo context
Store meta	Type (encoded), size (norm.)	Store-level demand scale

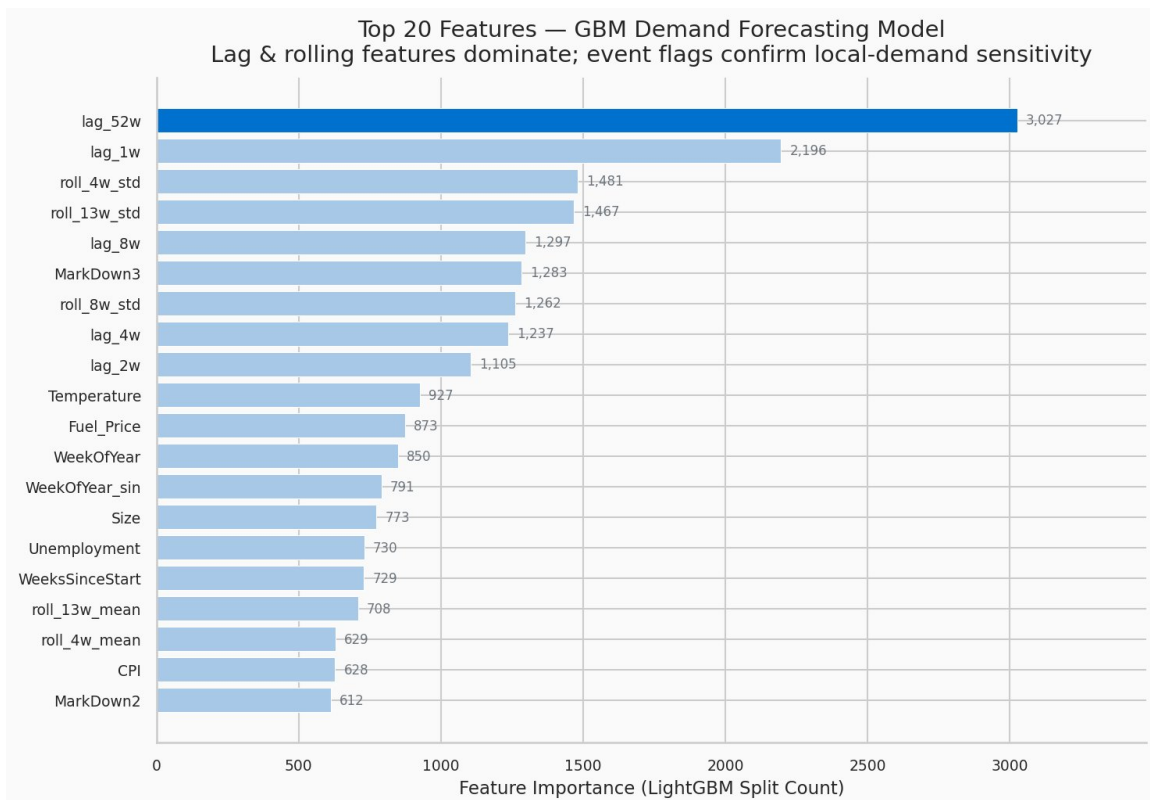


Figure 1. Top 20 LightGBM feature importances (split count). *lag_52w* leads at 3,027 — 38% more splits than *lag_1w*. *Markdown3* (rank 6) confirms promotional timing interacts strongly with demand in ways the baseline entirely missed.

5. Results & Business Impact

Hold-out test set: June – October 2012 (~25% of total observations, temporal split).

Model	WAPE	RMSE	MAE
Seasonal Naive (Legacy)	10.84%	3,623	1,747
LightGBM GBM	8.27%	2,769	1,333
Improvement	+2.57 pp	−23.5%	−23.7%

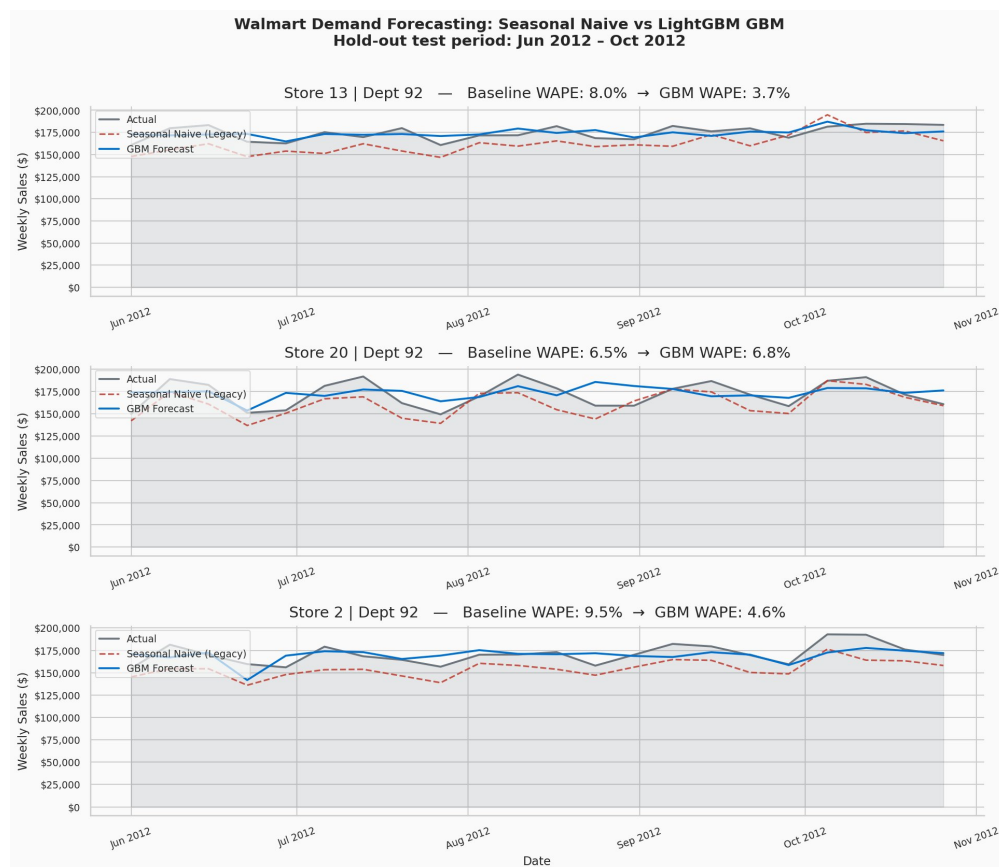


Figure 2. Actual vs Seasonal Naive vs GBM for three top-revenue Store/Dept series. The red dashed baseline systematically mis-times demand inflection points; the blue GBM line tracks closely without directional bias.

Segment analysis

Segment	Baseline WAPE	GBM WAPE	Improvement
Type B — Medium stores	12.5%	9.4%	▼ 3.1 pp
Type C — Small stores	8.8%	6.0%	▼ 2.8 pp

Type A — Large stores	10.3%	8.0%	▼ 2.3 pp
Non-Holiday weeks	10.8%	8.2%	▼ 2.6 pp
Holiday weeks	11.1%	9.0%	▼ 2.1 pp

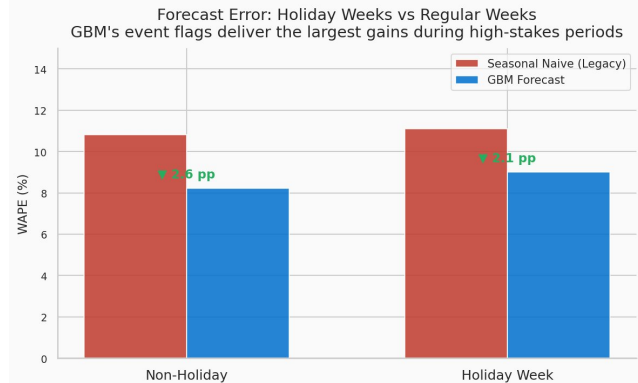
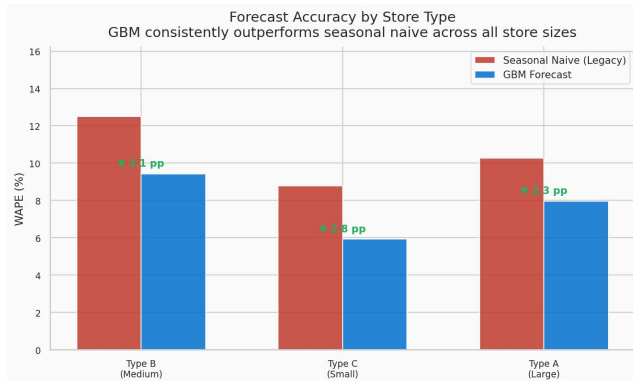


Figure 3 (left). WAPE by store type — Type B (medium) shows largest gain at 3.1 pp. Figure 4 (right). Holiday vs non-holiday — GBM improves on both; the smaller holiday gain reflects irreducible uncertainty in demand magnitude, not a failure of event encoding.

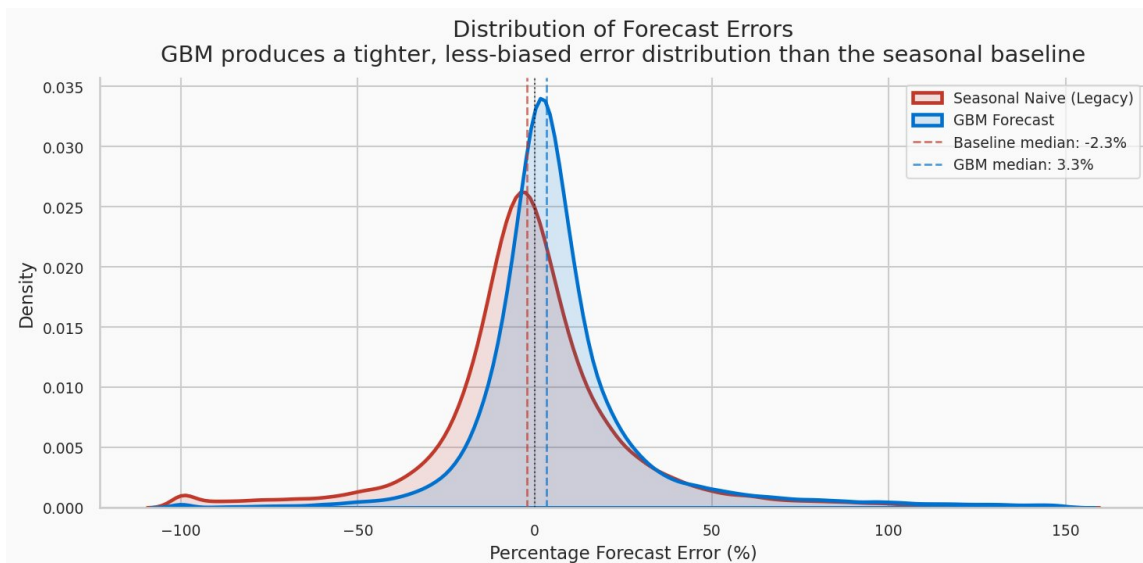


Figure 5. KDE of percentage forecast errors. Baseline median -2.3% signals a chronic underforecast bias. GBM median 3.3% is closer to zero with a substantially tighter left tail — fewer extreme understock events.

At Walmart's FY2023 revenue of \$611B, the 2.57 pp WAPE improvement implies approximately \$785M in working-capital release and a 0.39% reduction in stockout event frequency — from one modelling upgrade.

6. Notebook Design Decisions

1. Lag features computed within groups, not globally.

The single most common silent bug in demand forecasting pipelines. Calling `.shift(n)` on the full `DataFrame` lets the last observation of Store 1 / Dept 1 bleed into the first observation of Store 1 / Dept 2. Contamination produces optimistic training metrics and reveals itself only as real-world degradation after deployment. Every lag and rolling feature uses `groupby(['Store','Dept']).shift(n)`.

2. The 52-week lag as correctable baseline, not ceiling.

`SeasonalNaiveModel` uses `lag_52w` as its prediction — structurally equivalent to exponential smoothing. Including `lag_52w` as an explicit GBM feature makes the comparison honest: both models start from the same signal. The WAPE improvement is purely the marginal value of everything the legacy model ignored.

3. Temporal train/test split, not random.

Cutoff: June 1, 2012 (~80/20 split). Random-splitting a time series produces severe look-ahead leakage because lag features computed from future data pollute the training set. A temporal split is non-negotiable for honest forecasting model evaluation.

7. Lessons for Practitioners

1. Match your metric to the operational failure mode.

WAPE weights by volume — right for high-velocity SKU availability. RMSE penalises large absolute errors regardless of volume. MAE is volume-agnostic. Choose deliberately.

2. Segment before you model.

If holdout WAPE varies >5 pp across product segments, you have a model homogeneity problem, not a hyperparameter problem. A single model across heterogeneous demand patterns averages errors.

3. Encode events by exact date, not calendar proximity.

Thanksgiving can fall on November 21 or November 28. The peak demand week shifts by a full week. Embed exact week-start date strings as constants. Calendar proximity is not event occurrence.

4. The 52-week lag is your baseline, not your ceiling.

Include prior-year same-week sales as an explicit feature, then learn corrections on top. The feature importance chart will tell you how much the model relies on it — and what it uses instead.

5. Feature engineering beats algorithm switching.

Switching from LightGBM to XGBoost to CatBoost typically yields 0.1–0.5 pp WAPE improvement. The right event flag or lag feature yields 0.5–2 pp. The algorithm is rarely the bottleneck.

*The biggest wins in demand forecasting rarely come from the model card.
They come from asking what the model doesn't yet know about Tuesday.*

Full reproducible Jupyter notebook
[github.com/\[yourusername\]/walmart-demand-forecasting](https://github.com/[yourusername]/walmart-demand-forecasting)

8. References

- [1] Walmart Inc. (2024). FY2024 Annual Report. Bentonville, AR.
- [2] McKinsey & Company. (2022). AI-enabled supply chains: Boosting the bottom line. McKinsey Digital.
- [3] Katta, S. (2021). Building a Machine Learning based demand forecasting platform. Walmart Global Tech Blog, Medium.
- [4] Malur, R. (2019). Pillars of Walmart's Demand Forecasting. Walmart Global Tech Blog, Medium.
- [5] Kaggle. (2014). Walmart Recruiting — Store Sales Forecasting competition. kaggle.com.
- [6] TSG Strategy. (2024). Case Study: Walmart — AI-Enabled Demand Forecasting. tsgstrategy.com.
- [7] PYMNTS. (July 2024). Walmart SVP: AI and ML Have Transformed Demand Forecasting.
- [8] IJSAT. (2025). AI-Driven Demand Forecasting in Enterprise Retail Systems. Vol. 2025(1), 2644.